



UNIVERSIDAD NACIONAL DE INGENIERÍA

Facultad de Ciencias

Escuela Profesional de Ciencia de la Computación

EXAMEN FINAL

Curso: Fundamentos de Programación CC112 - A

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Código: 20232597K

1.1

```
daniel@Daniel: /mnt/e/UNI/I1/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danielldiaz-maker/Semana8/EF/Me
jora_Examen$ make valgrind
valgrind --leak-check=full --show-leak-kinds=all ./build/debug/Pregunta01
==58908== Memcheck, a memory error detector
==58908== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==58908== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==58908== Command: ./build/debug/Pregunta01
==58908==
Ingrese el numero de filas: 2
Ingrese el numero de columnas: 5
El vector suma filas es: 15 30
==58908==
==58908== HEAP SUMMARY:
==58908==    in use at exit: 0 bytes in 0 blocks
==58908==   total heap usage: 7 allocs, 7 frees, 75,840 bytes allocated
==58908==
==58908== All heap blocks were freed -- no leaks are possible
==58908==
==58908== For lists of detected and suppressed errors, rerun with: -s
==58908== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
daniel@Daniel: /mnt/e/UNI/I1/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danielldiaz-maker/Semana8/EF/Me
jora_Examen$
```

1.2

```
daniel@Daniel: /mnt/e/UNI/I1/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danielldiaz-maker/Semana8/EF/Me
jora_Examen$ make valgrind
valgrind --leak-check=full --show-leak-kinds=all ./build/debug/Pregunta01
==59054== Memcheck, a memory error detector
==59054== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==59054== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==59054== Command: ./build/debug/Pregunta01
==59054==
Ingrese el numero de filas: 6
Ingrese el numero de columnas: 2
El vector suma filas es: 3 6 9 12 15 18
==59054==
==59054== HEAP SUMMARY:
==59054==    in use at exit: 0 bytes in 0 blocks
==59054==   total heap usage: 11 allocs, 11 frees, 75,896 bytes allocated
==59054==
==59054== All heap blocks were freed -- no leaks are possible
==59054==
==59054== For lists of detected and suppressed errors, rerun with: -s
==59054== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
daniel@Daniel: /mnt/e/UNI/I1/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danielldiaz-maker/Semana8/EF/Me
jora_Examen$
```

2.1

```
daniel@Daniel: /mnt/e/UNI/I1/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danielldiaz-maker/Semana8/EF/Me
jora_Examen$ valgrind ./build/debug/Pregunta02
==59277== Memcheck, a memory error detector
==59277== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==59277== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==59277== Command: ./build/debug/Pregunta02
==59277==
Leyendo P1:
Digite el punto en x: 3
Digite el punto en y: 4
Leyendo P2:
Digite el punto en x: -3
Digite el punto en y: -5
Las coordenadas del puntero mas lejano son:
x: -3
y: -5
==59277==
==59277== HEAP SUMMARY:
==59277==    in use at exit: 0 bytes in 0 blocks
==59277==   total heap usage: 7 allocs, 7 frees, 76,004 bytes allocated
==59277==
==59277== All heap blocks were freed -- no leaks are possible
==59277==
==59277== For lists of detected and suppressed errors, rerun with: -s
==59277== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

2.2

```
daniel@Daniel: /mnt/e/UNI/II/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danielldiaz-maker/Semana8/EF/Mejora_Examen$ valgrind ./build/debug/Pregunta02
==59447== Memcheck, a memory error detector
==59447== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==59447== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==59447== Command: ./build/debug/Pregunta02
==59447==
Leyendo P1:
Digite el punto en x: 6
Digite el punto en y: 7
Leyendo P2:
Digite el punto en x: -8
Digite el punto en y: -12
Las coordenadas del puntero mas lejano son:
x: -8
y: -12
==59447==
==59447== HEAP SUMMARY:
==59447==   in use at exit: 0 bytes in 0 blocks
==59447==   total heap usage: 7 allocs, 7 frees, 76,004 bytes allocated
==59447==
==59447== All heap blocks were freed -- no leaks are possible
==59447==
==59447== For lists of detected and suppressed errors, rerun with: -s
==59447== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
daniel@Daniel: /mnt/e/UNI/II/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danielldiaz-maker/Semana8/EF/Mejora_Examen$
```

3.1

```
daniel@Daniel: /mnt/e/UNI/II/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danielldiaz-maker/Semana8/EF/Mejora_Examen$ valgrind ./build/debug/Pregunta03
==75698== Memcheck, a memory error detector
==75698== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==75698== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==75698== Command: ./build/debug/Pregunta03
==75698==

2. Agregar |1. Mostrar |0. Salir: 2
Digite el id: 12
Digite el Titulo del Proyecto: Puente Lomas
Digite el Presupuesto del Proyecto: 1200000
Digite la duracion del Proyecto en meses: 18

2. Agregar |1. Mostrar |0. Salir: 2
Digite el id: 12
ERROR: Ya existe otro proyecto con igual id.

2. Agregar |1. Mostrar |0. Salir: 2
Digite el id: 1
Digite el Titulo del Proyecto: Viaducto los reyes
Digite el Presupuesto del Proyecto: 3000000
Digite la duracion del Proyecto en meses: 20

2. Agregar |1. Mostrar |0. Salir: 1
LISTA DE PROYECTOS:
ID: 1 | Titulo: Viaducto los reyes | Presupuesto: S/3e+06 | Duracion: 20 meses
ID: 12 | Titulo: Puente Lomas | Presupuesto: S/1.2e+06 | Duracion: 18 meses

2. Agregar |1. Mostrar |0. Salir: 0
==75698==
==75698== HEAP SUMMARY:
==75698==   in use at exit: 0 bytes in 0 blocks
==75698==   total heap usage: 10 allocs, 10 frees, 77,498 bytes allocated
==75698==
==75698== All heap blocks were freed -- no leaks are possible
==75698==
==75698== For lists of detected and suppressed errors, rerun with: -s
==75698== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
daniel@Daniel: /mnt/e/UNI/II/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danielldiaz-maker/Semana8/EF/Mejora_Examen$
```

3.2

```
daniel@Daniel: /mnt/e/UNI/II/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danielldiaz-maker/Semana8/EF/Mejora_Examen$ valgrind ./build/debug/Pregunta03
==75897== Memcheck, a memory error detector
==75897== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==75897== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==75897== Command: ./build/debug/Pregunta03
==75897==

2. Agregar |1. Mostrar |0. Salir: 1
LISTA DE PROYECTOS:

2. Agregar |1. Mostrar |0. Salir: 2
Digite el id: 0
Digite el Titulo del Proyecto: Palacio de la Juventud
Digite el Presupuesto del Proyecto: 2500000
Digite la duracion del Proyecto en meses: 30

2. Agregar |1. Mostrar |0. Salir: 0
==75897==
==75897== HEAP SUMMARY:
==75897==   in use at exit: 0 bytes in 0 blocks
==75897==   total heap usage: 7 allocs, 7 frees, 76,841 bytes allocated
==75897==
==75897== All heap blocks were freed -- no leaks are possible
==75897==
==75897== For lists of detected and suppressed errors, rerun with: -s
==75897== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
daniel@Daniel: /mnt/e/UNI/II/VERANO/FUNDAMENTOS/CARPETA PARA REPOSITORIO UNI/pc02-cc112a-danielldiaz-maker/Semana8/EF/Mejora_Examen$
```

4.

```
daniel@Daniel: /mnt/e/UNI/11/SEMANO/FUNDAMENTOS/CARPETA PARA REPOSICION UNI/pc02-cc112a-danielldiaz-maker/Semana8/EF/Me
lora Examen$ valgrind ./build/debug/Pregunta04
==93537== Memcheck, a memory error detector
==93537== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==93537== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==93537== Command: ./build/debug/Pregunta04
==93537==

1.Agregar Circular 2.Agregar Rectangular 3.Mostrar Todos 4.Area Total 5.Detectar Punto 6.Mayor Cobertura 0.Salir
Op: 3

1.Agregar Circular 2.Agregar Rectangular 3.Mostrar Todos 4.Area Total 5.Detectar Punto 6.Mayor Cobertura 0.Salir
Op: 1
ID, X, Y, Radio: 12 2 3 5

1.Agregar Circular 2.Agregar Rectangular 3.Mostrar Todos 4.Area Total 5.Detectar Punto 6.Mayor Cobertura 0.Salir
Op: 2
ID, X, Y, Ancho, Alto: 10 4 6 8 5

1.Agregar Circular 2.Agregar Rectangular 3.Mostrar Todos 4.Area Total 5.Detectar Punto 6.Mayor Cobertura 0.Salir
Op: 3
[Circular] ID: 12 | Pos: (2,3) | Radio: 5 | Area: 78.5397
[Rectangular] ID: 10 | Pos: (4,6) | Dim: 8x5 | Area: 40

1.Agregar Circular 2.Agregar Rectangular 3.Mostrar Todos 4.Area Total 5.Detectar Punto 6.Mayor Cobertura 0.Salir
Op: 4
Area Total: 118.54

1.Agregar Circular 2.Agregar Rectangular 3.Mostrar Todos 4.Area Total 5.Detectar Punto 6.Mayor Cobertura 0.Salir
Op: 5
Punto X Y: 2 3
Detectado por 1 sensor(es)

1.Agregar Circular 2.Agregar Rectangular 3.Mostrar Todos 4.Area Total 5.Detectar Punto 6.Mayor Cobertura 0.Salir
Op: 6
Mayor:
[Circular] ID: 12 | Pos: (2,3) | Radio: 5 | Area: 78.5397

1.Agregar Circular 2.Agregar Rectangular 3.Mostrar Todos 4.Area Total 5.Detectar Punto 6.Mayor Cobertura 0.Salir
Op: 0
==93537==
==93537== HEAP SUMMARY:
==93537==    in use at exit: 0 bytes in 0 blocks
==93537==   total heap usage: 17 allocs, 17 frees, 76,457 bytes allocated
==93537==
==93537== All heap blocks were freed -- no leaks are possible
==93537==
==93537== For lists of detected and suppressed errors, rerun with: -s
==93537== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
daniel@Daniel: /mnt/e/UNI/11/SEMANO/FUNDAMENTOS/CARPETA PARA REPOSICION UNI/pc02-cc112a-danielldiaz-maker/Semana8/EF/Me
```